Terrain Aware Wheelchair Navigation in Outdoor Spaces Using Hybrid Fuzzy A* Algorithm

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Abstract

Wheelchair users often face difficulty to get around outside on unpaved surfaces because of the different types of terrain, changes in slope, and safety risks. A* and other common path-finding algorithms usually use simple weighted cost functions, which don't show the subtlety and complexity of moving through the real world. This paper presents a comparative analysis between a Weighted A* algorithm and a novel Hybrid A* algorithm driven by a Fuzzy Inference System. Both algorithms were tested in a procedurally generated 3D simulation environment with defined terrain types and slopes. The results show that the Weighted A* algorithm may find a shorter path; however, the resulting paths are often impractical or unsafe in reality. In contrast, the Fuzzy A* hybrid algorithm generates a more human-like path, which is safer to travel.

Keywords: Hybrid fuzzy A* algorithm, obstacles, path planning, terrain, weighted a* algorithm, wheelchair

Introduction

Navigating the outdoor environment still poses a challenge for wheelchair users due to unpredictability of terrain. Unlike the flat surfaces in indoor or outdoor environments, outdoor areas like parks or even pathways in the countryside have grass, gravel, or dirt surfaces that provide different levels of traction and vibrations, in addition to the presence of slopes that complicate the navigation of the wheelchair. For example, a wheelchair may be stable on a slope on a paved surface but not on grass or dirt surfaces, leading to problems of stability, tilting, or even getting stuck in the terrain, which cannot be addressed by conventional path-planning algorithms. Conventional algorithms, such as the Weighted A* algorithm, determine the optimality of a path by the linear cost function that considers the distance, slope, and terrain of the path. Unfortunately, this strict additive structure fails to reflect the subtle and often dependent nature of real-world wheelchair motion. For instance, a moderate slope on unstable terrain, is exponentially more dangerous than each of those factors in isolation. Therefore, although Weighted A* may indeed yield mathematically shorter paths, they are often unsafe or undesirable paths in practice.To address the limitations of the conventional algorithms, a Hybrid A* algorithm that uses a Mamdani-type Fuzzy Inference System (FIS) has been proposed that uses fuzzy logic to mimic the ability of a human to assess the cost of navigating a terrain by considering the slope, terrain, and distance in a more realistic manner than the conventional Weighted A*.

Literature review

Wang et al. [1] proposed a genetic algorithm (GA) that ensures safety in trajectory planning by penalizing cliffs and slopes using a specific evaluation function. Although this solution improves stability and safety over A*, it results in longer trajectories that require more computational time.

Y.Li et al. [2] proposed framework for multilevel path planning in real-time, considers coarse path planning and local re-planning through hierarchical search. It improves efficiency in large environments, taking into account terrain roughness and slope. The framework takes into account dynamic re-planning, but it doesn't use dynamic reasoning; instead, it uses static cost assumptions.

Zhou et al. [3] examine the slope cost function. This cost function gives an idea of how likely it is that someone will slip and adds a penalty for how slippery the ground is. A better A* algorithm with slope weighting is more stable and keeps dangerous situations to a minimum. But the cost structure can't be modified on immediate basis based on rules about behaviour or context.

Gunathilaka et al. [4] propose a path planner that integrates A* with image-based semantic segmentation to account for the terrain's semantics. Surface labels like road, grass, gravel, and sand help figure out the cost based on risk and visibility so that people can choose safer paths. But relying on visual classification can make things worse in bad light or rough terrain.

Saranya et al. in [5] try to overcome the limitations of A* in elevation-based terrain navigation by a risk model that considers slope and roughness. Cells are ranked by stability and traction to improve safety. However, this increases computational cost and does not employ integrated safety reasoning.

Wen Bao et al. in [6], A* algorithm is improved by using a 16-direction search, dynamic weighting, and turn optimization to generate smoother paths. However, these algorithms do not account for safety, comfort, or terrain considerations. A hybrid fuzzy A* algorithm was developed to handle uncertainties in order to ensure safe wheelchair movements on uneven terrain.

Sandeep et al [7] utilized fuzzy logic to analyze various combinations of membership functions for mobile robot path planning to improve both navigation accuracy and obstacle detection/avoidance.

Meghana et al [8] presented a fuzzy inference system that can be dynamically tuned using the dynamic window approach to allow for better real-time trajectory selection.

Divya et al [9] used Type-1 and neuro fuzzy models along with machine learning techniques to efficiently and adaptively provision 5G resources in networks with varying external conditions.

Jasna et al. in their paper [10] proposed a modified real-time A* algorithm for path planning in static environments. In the proposed algorithm, the cost function is improved by adding the position, heading, and node weights of the obstacles.

Existing research has been able to advance terrain-aware path planning for robots and vehicles through the application of cost-based safety strategies. However, the strategies have not been effective in the consideration of human comfort factors, reasoning, and the interpretation of terrain safety using fuzzy reasoning. To overcome the shortcomings of the existing strategies, the Hybrid Fuzzy A* strategy has been proposed.

Methodology

Simulation Environment Generation

Terrain Generation: A 150*150 grid-based 3D terrain was procedurally generated to serve as the test environment. A realistic rough ground surface was created using the superposition of multiple scales of Perlin-like noise. It is generated by applying Gaussian filters to random noise fields. This surface was then normalized and contoured to create a mix of flat lowlands and rougher hills using MATLAB.

Terrain Cost Zones: The terrain was divided into three zones.
• The base terrain represents a grass-like surface rendered in a blue-green winter colourmap with the cost defined as 25.
• The yellow zones on the terrain represent a pavement-like surface with the cost defined as 10.
• The red zones on the terrain represent a gravel-like surface with the cost defined as 80.

Obstacle Placement: The terrain was then populated with a couple of solid, impassable obstacles, which were placed randomly.

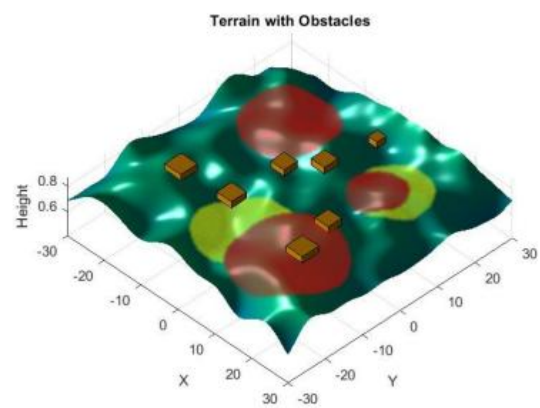

Figure 1 3D terrain model with obstacles
Source: Generated from MATLAB

Pathfinding Algorithm Implementation - Weighted A* Algorithm

Weighted A* and Hybrid A* Fuzzy algorithms were implemented to traverse this environment from a random start point to a random goal point. The weighted A* Algorithm is a modification of the A* Algorithm that increases the speed of the search by assigning more weight to the heuristic (h-cost), thereby making it more directed towards the goal. The motivation is to reach the goal as fast as possible, which usually decreases runtime, but it might sometimes be at the expense of safety or feasibility of the path, since the algorithm may choose to take a slightly riskier new route to get to the goal sooner rather than a safer route that is slightly longer. The algorithm was tailored to navigate outdoor wheelchair travel by creating a movement cost function that integrates distance, difficulty of the terrain, and severity of slope based on adjustable weights. Elevation data was used to classify slope conditions and apply penalties or rewards, while a cost-map identified terrain type. Risks of tipping over on unsafe slopes, as well as obstacles, were eliminated using the tipping-risk thresholds. The heuristic was computed on the basis of minimizing the cost-per-unit movement to enhance convergence. The algorithm expanded nodes according to the lowest weighted cost until the goal was reached, reconstructing the path through stored parent references. Mathematically, the total cost function f(n) for a node n is defined as [6]:

$$f(n) = g(n) + h(n) \qquad (1)$$

where g(n) is the actual cost from the start node to node n, and h(n) is the heuristic estimate of the cost from node n to the goal.

The cost to move from a current node n to a neighboring node n' is denoted as c(n,n'). It is calculated as a weighted sum of three environmental factors: Euclidean distance, slope change and terrain surface type. The cost function is defined as:

$$c(n,n') = wd \cdot Cdist + ws \cdot Cslope + wt \cdot Cterrain \qquad (2)$$

where the weights are experimentally set to
$w_d = ws = w_t = 0.333$

Distance Component: The distance term penalizes longer transitions, and it is calculated as the Euclidean Norm.

$$D(n,n') = || n - n' ||_2 \qquad (3)$$

Slope Component: Let H denote the terrain elevation map. The local slope is defined as:

$$\Delta H = H(n') - H(n) \qquad (4)$$

The slope cost includes asymmetric penalties for inclines and declines.

$$S(n,n') = \begin{cases} 1 + \Delta H \cdot P_{steep-inc}, & \Delta H > \tau_{inc,} \\ 1 + \Delta H \cdot P_{gentle-inc}, & 0 < \Delta H \le \tau_{inc,} \\ 1 \cdot R_{gentle-dec}, & \tau_{dec} < \Delta H \le 0, \\ 1 + |\Delta H| \cdot P_{steep-dec}, & \Delta H \le \tau_{dec.} \end{cases} \qquad (5)$$

where:

- $P_{steep-inc} = 150$
- $P_{gentle-inc} = 30$
- $P_{steep-dec} = 50$
- $R_{gentle-dec} = 0.5$

Incline threshold:
- $\tau_{inc} = 0.08$

Decline threshold:
- $\tau_{dec} = -0.1$

Terrain Component: The terrain type cost is calculated as follows: (6)

$$T(n') \in \{10\ (pavement), 25\ (ground), 80\ (gravel)\}$$

Penalty coefficients, slope thresholds, terrain cost values, and weight factors were each iteratively evaluated via simulation in various terrain incline, decline, and mixed cases. The final parameters avoid steep slopes and difficult surfaces yet provide stable, reasonable mobility for a smart wheelchair.

A* Update and Heuristic: The g-score is then calculated as:

$$g(n') = g(n) + C(n,n') \qquad (7)$$

The heuristic is computed as:

$$h(n') = \| n' - goal ||_2 \cdot C_{min} \qquad (8)$$

where C_{min} is the minimum possible per-step obtained by:

$$C_{min} = w_d Dmin + w_s Smin + w_t Tmin \qquad (9)$$

Thus, the Weighted A* Algorithm cost function becomes:

$$f(n') = g(n') + h(n') \qquad (10)$$

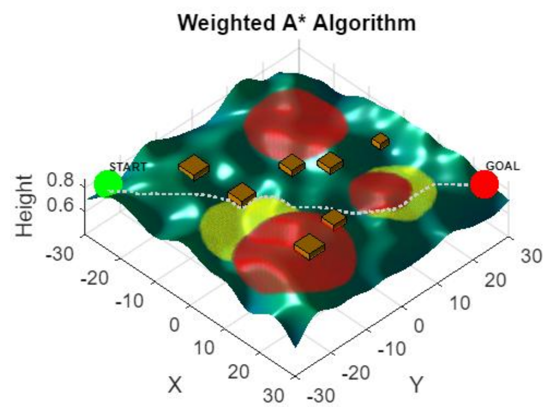

Figure 2 Path traced by Weighted A* Algorithm
Source: Generated from MATLAB

While this method discovers a quicker and more direct path, it can find paths that are not the safest choice in actual terrain, which points to why additional refinement is beneficial to safety-critical mobility systems.

Hybrid Fuzzy A* Algorithm

Fuzzy logic is a soft computing method that allows reasoning with uncertainty by expressing degrees of truth rather than representing true or false. Unlike traditional crisp methods, fuzzy inference deals well with vague, imprecise, or nonlinear characteristics of the environment, which makes it ideal for use in real-world problems like outdoor mobility, where terrain and slopes cannot be classified into rigid categories. Fuzzy sets are commonly implemented using membership functions. The two most widely used forms are as follows:

Triangular Membership Function (trimf):

- It is defined by a triplet (a, b, c) which forms a triangle with peak at b.
- It is used for smooth and symmetric fuzzy categories.

Trapezoidal Membership Function (trapmf):

- It is defined by four parameters (a, b, c, d) which produces a plateau where the membership equals 1.
- It is suitable for broad or less sharply defined fuzzy categories.

The fuzzy inference system can use these membership functions to convert physical values, such as a slope or the cost of the terrain into interpretable linguistic terms.

The idea for combining fuzzy logic with the A* algorithm comes from the limitations observed in the Weighted A* algorithm. While Weighted A* uses heuristic distance to find shorter and faster paths, it tends to take dangerous paths over less dangerous paths that may take more time and distance, especially on steep slopes or rough ground. A hybrid Fuzzy A* algorithm was proposed, which keeps the basic A* search structure, but uses a fuzzy inference system (FIS) to evaluate its movement costs and rewards to account for slopes and surface types and movement directions, as a way to account for uncertainty. The proposed navigation system contains three inputs and one output, all of which are described using fuzzy sets. The input and output of the new system are described below, together with the membership functions.

Fuzzy Input Variable- Slope: The slope represents the elevation difference between two adjacent nodes, normalized within the interval [−0.25, 0.25]. Five fuzzy sets are used to represent different slope behaviors.

Table 1: Fuzzy Sets for Slope Input Variable

Linguistic Term	MF Definition	Interpretation
Steep Decline	trapmf: [−0.25, −0.25, −0.15, −0.08]	Rapid downward slope; dangerous
Gentle Decline	trimf: [−0.12, −0.05, 0]	Mild downhill movement
Flat	trimf: [−0.04, 0, 0.04]	Level surface
Gentle Incline	trimf: [0, 0.06, 0.12]	Mild uphill slope
Gentle Incline	trapmf: [0.08, 0.15, 0.25, 0.25]	Highly risky uphill.

Fuzzy Input Variable-Terrain Type: Terrain difficulty is mapped into the normalized range [0, 100], which represents pavement, ground, and gravel surfaces. Three fuzzy sets are defined to model them.

Table 2: Fuzzy Sets for Terrain Type Input Variable

Linguistic Term	MF Definition	Interpretation
Smooth	trapmf: [0, 0, 5, 15]	Pavement or extremely smooth regions.
Moderate	trimf: [10, 25, 40]	Regular ground or mixed terrain.
Rough	trapmf: [35, 50, 100, 100]	Gravel zones; unstable and high-cost.

Fuzzy Input Variable: Distance: Distance discriminates between straight and diagonal movements in the grid. Two fuzzy sets are defined to represent the cost characteristics associated with orthogonal and diagonal transitions.

Table 3: Fuzzy Sets for Distance Input Variable

Linguistic Term	MF Definition	Interpretation
Straight	trimf: [0.8, 1.0, 1.2]	Cost associated with orthogonal moves.
Diagonal	trimf: [1.2, 1.414, 1.6]	Longer diagonal motions.

Fuzzy Output Variable: Movement Cost: The output variable spans the range [0, 200] and represents the movement cost assigned to each transition. Five fuzzy sets are defined to characterize varying levels of traversal difficulty.

Table 4: Fuzzy Sets for Movement Cost Output Variable

Linguistic Term	MF Definition	Meaning
Very Low	trimf: [0, 5, 10]	Ideal movement; safe and cheap.
Low	trimf: [5, 15, 25]	Minor effort required
Medium	trimf: [20, 50, 80]	Moderately difficult region
High	trimf: [70, 100, 130]	Challenging or potentially risky
Very High	trapmf: [120, 160, 200, 200]	Extremely unsafe - steep or rough terrain.

The Movement Cost for the node extension will be incorporated into the A* structure via the updated f(n) = g(n) + h(n) formulation.

Ri: IF S is Ai AND T is Bi AND D is Ci, THEN C_{move} is Z (11)

where:

• Ai, Bi, Ci, Zi - fuzzy sets (linguistic labels)

• S, T, and D represent the slope, terrain, and distance input

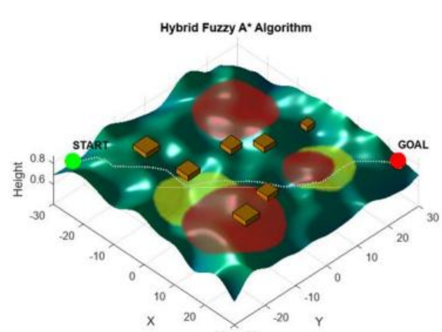

Figure 3 Path traced by Hybrid Fuzzy A* Algorithm

Source: Generated from MATLAB

The resulting Hybrid Fuzzy A* model produces a softer, safety-conscious path and avoids hard decision boundaries, allowing the algorithm to operate realistically in a challenging environment. Whereas Weighted A* can take shorter, unsafe paths, Hybrid Fuzzy A* will consistently identify a smoother and safer path, balanced against effective goal convergence, making it the best out of the two for navigating a wheelchair over uneven outdoor terrain.

Key Performance Indicators

The performance of the proposed Hybrid Fuzzy A* and Weighted A* algorithms was evaluated using a set of performance metrics as follows:

Compute Time(seconds)

It represents the total execution time required to generate a path. The Hybrid Fuzzy A* Algorithm resulted in decreased planning time which illustrates increased efficiency.

Path Length (Nodes)

It measures the total number of discrete grid nodes used along the solution path. The weighted A* expanded fewer nodes as it prioritises paths with shorter distances.

Path Distance

The measurement calculates all Euclidean distance which a person travels throughout the entire path. The Weighted A* Algorithm created a riskier but more direct route through its design which brought users close to nearby hazards.

Average Slope Δ

The measurement shows average elevation changes which occur between different steps of the measurement process. The wheelchair achieves better stability through its decreased slope variation which leads to lower tipping hazards because the Hybrid Fuzzy A* algorithm created the least slope alterations which proved to be its safety advantage.

Table 5: Performance Comparison Between Weighted A* and Hybrid Fuzzy A* Algorithm

Metric	Weighted A*	Hybrid Fuzzy A*
Compute Time(s)	180.2928	114.4468
Path Length (Nodes)	162	185
Path Distance	204.08	222.52
AVG Slope Δ	0.0021	0.0018

Results

The Hybrid Fuzzy A* Algorithm shows superior performance to the Weighted A* Algorithm according to performance indicators which evaluate wheelchair safety on uneven ground. The Hybrid Fuzzy A* Algorithm provides a path which offers better safety and better adaptability to various types of terrain compared to the Weighted A* Algorithm which aims to create shorter paths through its distance minimization method. The system demonstrates improved node evaluation efficiency through its ability to decrease computing time by a substantial amount. The Hybrid Fuzzy A* algorithm demonstrates better stability through its lower average slope variation compared to the Weighted A* algorithm which provides shorter distances thus making it easier for wheelchair users to navigate.

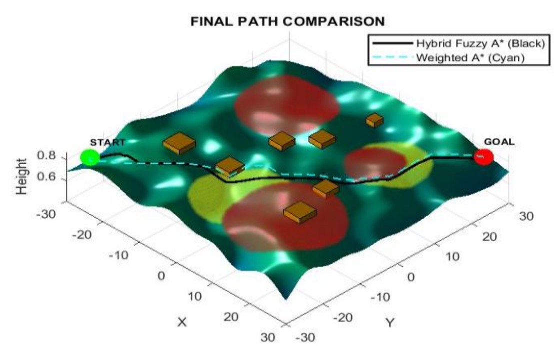

Figure 4 Path Comparison between A* and Hybrid Fuzzy A*
Source: Generated from MATLAB

Visual comparisons show that Weighted A* approaches obstacle boundaries at an unsafe distance which Hybrid Fuzzy A* uses smart deviation to maintain safe operation. The Hybrid Fuzzy A* algorithm outperforms other methods because it evaluates both path efficiency and real-world safety limits of actual terrain.

Significance

The research study develops an intelligent path-planning system which solves the main problem of autonomous outdoor wheelchair mobility on uneven surfaces. The study demonstrates its value because it enables users to gain more independence while preventing accidents and improving operations in real-world situations which occur in unstructured spaces.

Novelty

The research introduces a new path planning method which combines fuzzy logic decision making with the traditional A* algorithm to create a solution for wheelchair movement on uneven outdoor surfaces. The traditional A* method depends entirely on numerical heuristics which fail to account for safety requirements. The new method uses a dynamic fuzzy inference system to calculate costs based on slope gradients and terrain difficulty and movement direction, which helps the user maintain stability and safety while traveling. The proposed hybrid Algorithm creates an effective navigation system which prioritizes safety for smart wheelchairs in outdoor settings.

Conclusion

The study introduced a hybrid Fuzzy A* path-planning algorithm which provides autonomous wheelchairs with safe and reliable navigation capabilities through complex outdoor environments. The proposed method combines traditional heuristic search techniques with fuzzy logic-based cost assessment to achieve an equilibrium between optimal vehicle driving paths and safety standards that protect against slope and terrain dangers. The comparison results between Weighted A* and Hybrid Fuzzy A* Algorithm demonstrate that Hybrid Fuzzy A* method effectively avoids dangerous terrain areas while maintaining better distance from obstacles and delivering smoother stable paths which enhance overall navigational strength.

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